

What is an RTOS?

A real time operating system (RTOS) is an OS that guarantees a certain capability within a specified time constraint. There are two types of RTOSs—hard and soft.

Consider an automated assembly line that makes use of robots for assembling a ceiling fan. The robot will be required to place the ball bearings of the fan at the precise moment when each fan reaches a specific position at a certain time interval. In a hard RTOS, if the calculation could not be performed for making the ball bearing available at the right time, the OS would terminate with a failure. For a soft RTOS, the assembly line would continue to operate even though the fan is missing ball bearings. This is bad because the productivity of the whole assembly line is lost. Generally used OSes like Microsoft's Windows 2000 or IBM's OS/390 can be considered as RTOSs since they satisfy some conditions of an RTOS. However, they do not meet all the requirements of an RTOS.

In general, an RTOS has the following features:

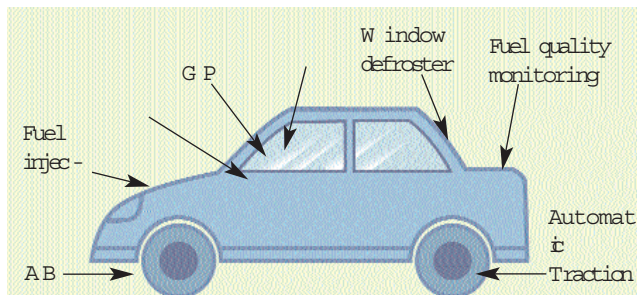
- Multitasking
- Process threads that can be prioritised
- A sufficient number of interrupt levels

RTOSs are usually employed in small embedded operating systems that are packaged as part of microdevices like microprocessors.

The major difference between embedded systems and RTOSs is that embedded systems are basically platform-less systems, that is, they do not use any operating system, whereas RTOSs work on specific OSes.

Most of the embedded systems in automobiles are rugged in nature, as most of these systems are made up of a single chip. Other factors aiding their use are the low costs involved, ease of development, and the fact that embedded devices can be networked to act as sub-modules in a larger system. No driver clashes or 'system busy' conditions happen in these systems. Their compact profiles enable them to fit easily under the cramped hood of a car.

Embedded systems can be used to implement features ranging from adjustment of the suspension to suit road conditions and the octane



Embedded systems in a car

content in the fuel to antilock braking systems (ABS) and security systems. Speaking of things nearer home, the 'computer chip' that controls fuel injection in a Hyundai Santro or the one that controls the activation of air bags in the recently launched Fiat Siena Weekend is nothing but an embedded system. Right from brakes to automatic traction control to air bags and fuel/air mixture controls, there may be up to 30-50 embedded systems within a present-day car. And this is just the beginning.

Home is where the chip is

How would you like to live in a house that recognises your fingerprints and opens the front door automatically? Imagine a fridge that checks your food supplies and places orders to replenish them; or intelligent lights and air conditioners that adjust their settings automatically according to the light and heat available at various times of the day. What about a microwave oven that can send a message to your PDA saying that the cake you kept to bake is done? These are all possible in the near future. In fact, many of these devices have already made their debut using embedded systems for controlling them and the Bluetooth technology for communication.

Homes of the future will employ embedded technology devices to become 'intelligent'. An intelligent house is essentially a network of smart devices, which could be a fridge, TV, microwave oven, washing machine or any of the numerous domestic appliances that we use at home. They will be able to optimise themselves or report maintenance problems to technicians using

built-in management functions. And they will also be connected to the Internet in most cases.

Real time OSes will work wonders when it comes to intelligent houses. The huge advantage they have over desktop OSes is their low cost. This factor alone can keep desktop OSes out of the running, but there are other reasons as well. An intelligent house may have more than 50 different devices in it. Imagine the cost of licensing that many copies of Windows 2000! Also, RTOSs are capable of being inter-networked, which enables co-ordinated functioning between devices at a scale that is unimaginable if desktop

OSes are used. Imagine getting scalded while having a hot shower—all because of an OS 'freeze' or 'crash'. Ouch!

The doctor will see you now

Imagine a time when body transplants like cardiac pacemakers will be able to monitor and manage themselves remotely. These systems will be so compact that the patient wouldn't even be aware that they are embedded in

his body. Another point in favour of RTOSs is their stability, instantaneous response and efficiency. No OS crashes or delays in responding, as is the case with most desktop OSes. After all, which patient can rest easy with the thought of his life literally hanging by a thread due to the time taken by the OS to load!

Embedded technology advances are pointing towards the use of pacemakers that can be transplanted in or near the heart itself. The pacemaker will be able to monitor parameters like blood pressure, blood flow, pulse rate, temperature, etc, using microsensors planted in various parts of the body. This capability will enable the pacemaker to automatically vary its operation to suit the changing body conditions. It will also transmit data using wireless transmission, thus enabling a doctor to constantly monitor its operation. In most cases, wireless transmission is likely to be done by a transmitter implanted near the surface of the skin. In case any abnormality is detected, the doctor will be able to take remedial action even from a remote location.

Wired wearables

A mobile phone in the form of a ring or a earring? What about cool sunglasses, with streaming video displays built into them? All these can soon be a reality. Embedded systems have a small footprint and consume very little power, which makes them ideal for wearable computing applications. The minimal system requirements of these devices ensure that the hardware is almost microscopic.

IBM is already working on the prototype of

